

CUSTOMER BEHAVIOR ANALYSIS

Segmentation, Purchase Patterns & Churn Risk

PREPARED FOR ALFIDO TECH

250,000 transactions · 49,661 customers · 2020-01-01 to 2023-09-13

TOTAL REVENUE

\$681M

AVG PURCHASES /
CUSTOMER

5.0

OVERALL CHURN

20%

Executive Summary

This report analyzes 250,000 e-commerce transactions from 49,661 customers to identify behavioral segments, purchase patterns, and churn risk for Alfido Tech. Customers were segmented using both rule-based RFM (Recency, Frequency, Monetary) scoring and K-Means clustering. Total revenue in the dataset is **\$681,346,299**, with an average of **5.0** purchases per customer and an overall churn rate of **20.0%**.

Key Takeaways

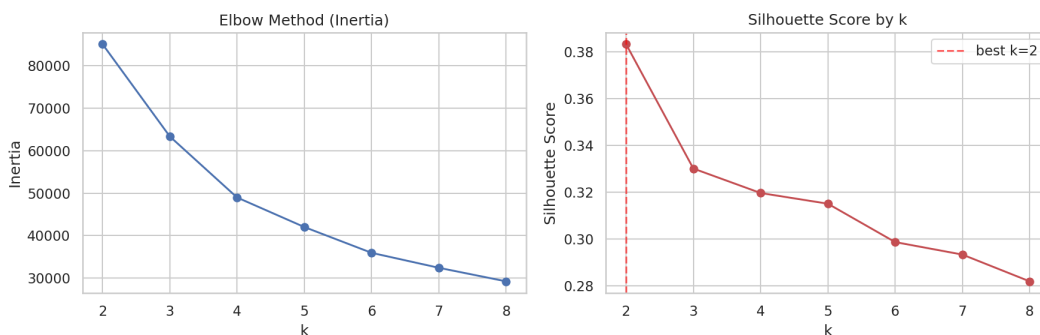
- Revenue is concentrated: the top segment (~32% of customers) generates ~47% of total revenue.
- Churn is broadly uniform (~20%) across every segment and RFM tier — it does not currently track behavioral engagement in this dataset.
- Category and payment preferences are evenly distributed across segments, with no single dominant category or payment method.
- Purchase volume and revenue are roughly flat across the observed period, pointing to acquisition — not seasonality — as the main growth lever.

Data Cleaning

- Loaded 250,000 raw transaction rows across 49,661 unique customers; parsed dates, filled missing return flags (0 = no return), dropped a redundant duplicate age column.
- No negative prices, quantities, totals, or exact duplicate rows found.
- Noted a data-generation artifact: Total Purchase Amount $\neq$ Product Price $\times$ Quantity for all rows — Total Purchase Amount was treated as the ground-truth revenue figure throughout.

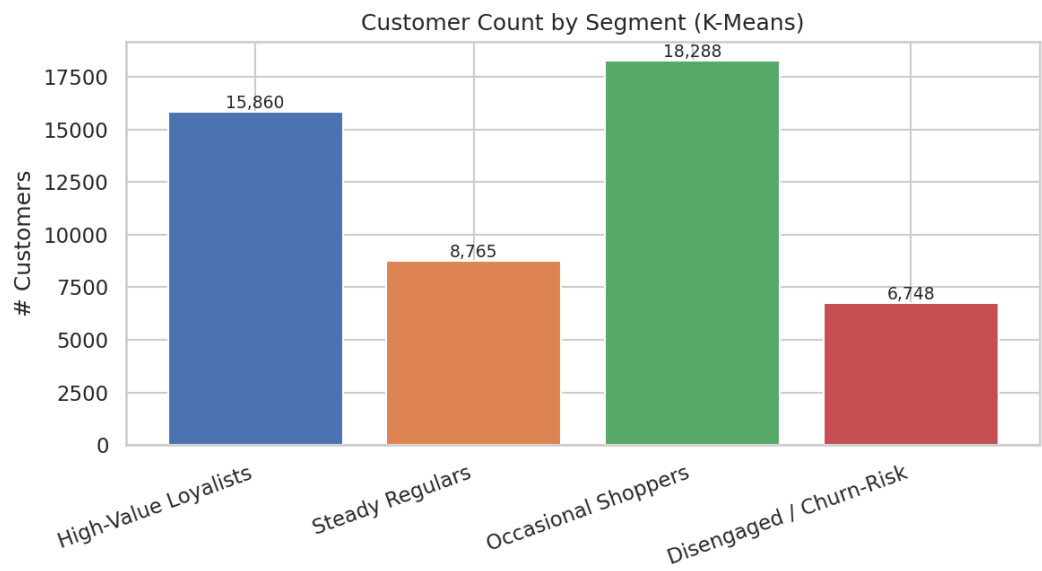
Segmentation Approach

Two complementary methods were used. A rule-based **RFM scoring** model assigns each customer a 1-5 quintile score on Recency, Frequency, and Monetary. Separately, **K-Means clustering** was applied to log-scaled, standardized RFM features to derive data-driven segments — cluster count ($k=4$) was chosen by comparing inertia and silhouette score across $k=2-8$, balancing statistical fit with business interpretability.



Elbow method and silhouette score used to select the number of clusters.

Customer Segments



Number of customers per K-Means segment.

Segment	Customers	% Base	Avg Recency	Avg Freq.	Avg Spend	% Revenue	Churn %
High-Value Loyalists	15,860	31.9%	202d	7.1	\$20,070	46.7%	20.2%
Steady Regulars	8,765	17.6%	28d	5.8	\$15,641	20.1%	20.4%
Occasional Shoppers	18,288	36.8%	334d	4.0	\$10,701	28.7%	19.7%
Disengaged / Churn-Risk	6,748	13.6%	514d	2.0	\$4,483	4.4%	19.9%

Table 1. Segment profile summary.

Segment Descriptions

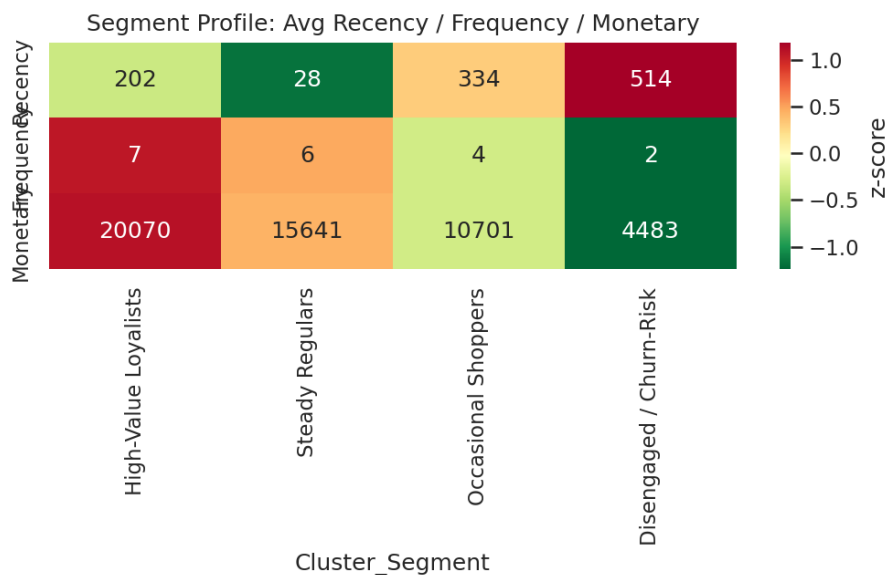
High-Value Loyalists (31.9% of customers) — Highest spend and frequency of any segment; nearly half of all revenue comes from this group despite being under a third of customers. Priority segment for retention investment.

Steady Regulars (17.6% of customers) — Most recently active segment with solid frequency and spend — dependable repeat buyers, good candidates for loyalty-program upsell.

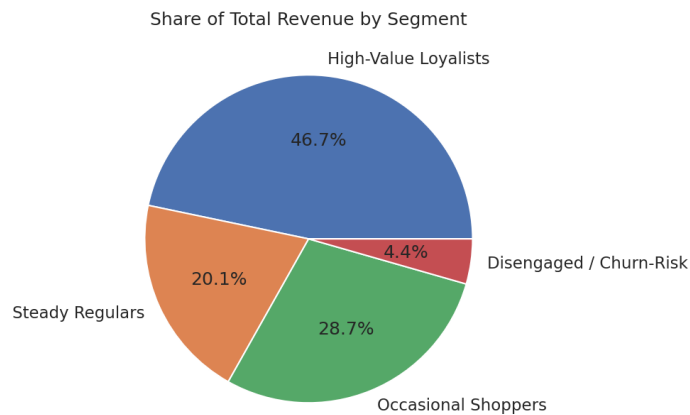
Occasional Shoppers (36.8% of customers) — The largest segment by customer count, with moderate spend but long gaps between purchases. Best opportunity for frequency-building campaigns.

Disengaged / Churn-Risk (13.6% of customers) — Lowest frequency and spend, with the longest average recency (500+ days). Smallest share of revenue; primary target for win-back offers.

Segment Profile & Revenue Concentration



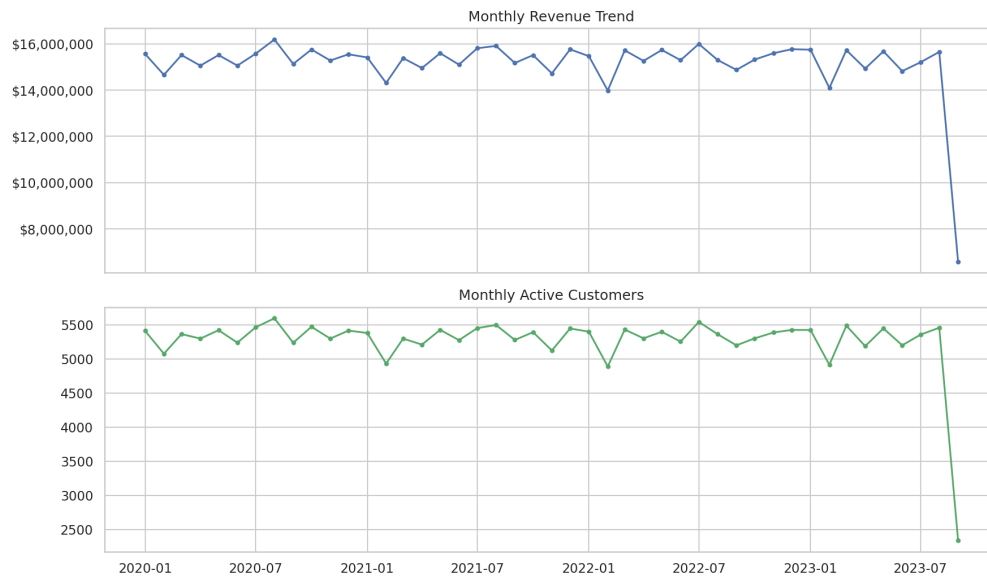
Relative Recency / Frequency / Monetary profile per segment.



Share of total revenue contributed by each segment.

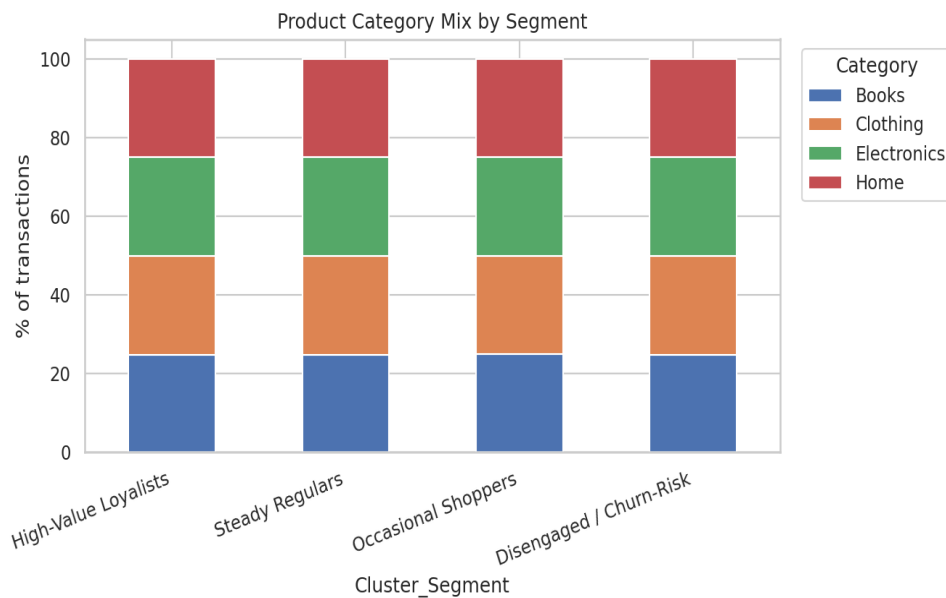
High-Value Loyalists alone generate **46.7%** of total revenue from just **31.9%** of customers — the clearest signal in this analysis for where retention investment should be prioritized.

Purchase Patterns & Category Mix



Monthly revenue and active-customer counts, 2020-2023.

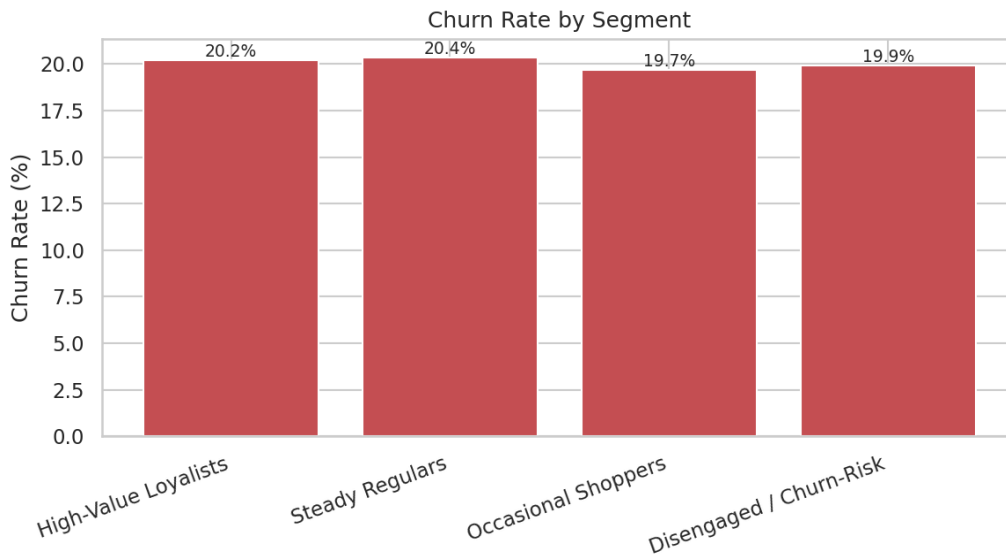
Revenue and active customer counts are broadly stable month-to-month, without strong seasonal peaks or a clear growth trend — suggesting new customer acquisition, not seasonality, is the primary lever for future growth.



Product category mix, by segment.

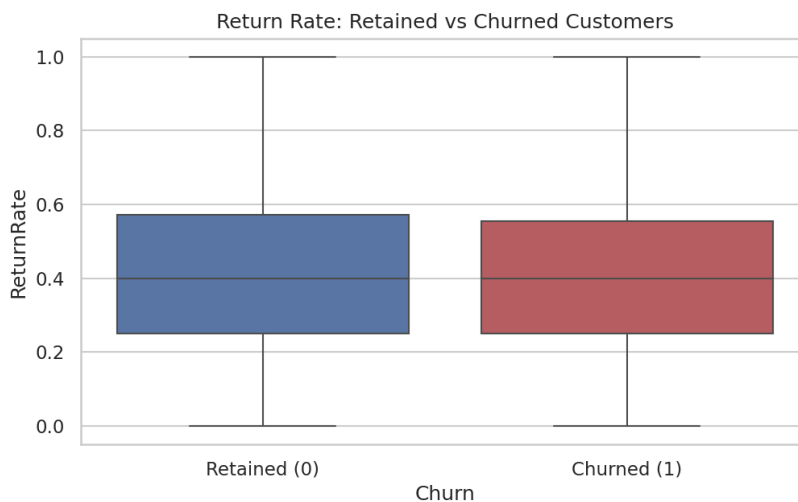
Category mix is nearly identical across all four segments (~25% each across Electronics, Clothing, Home, and Books) — category preference is an individual-level signal, not a segment-level one.

Churn Risk — A Counterintuitive Finding



Churn rate by segment.

Churn holds steady at roughly **20%** across every segment and RFM tier. In a typical e-commerce dataset, churn concentrates in low-recency, low-frequency customers — its absence here means the churn label behaves independently of purchase behavior.



Return rate distribution for retained vs. churned customers.

Return behavior is also similar between retained and churned customers. Any churn-prevention program built on these segments should be validated against live retention outcomes once deployed, rather than assumed from this label alone.

Recommendations for Alfido Tech

1. Protect the High-Value Loyalists segment first.

This ~32% of customers generates ~47% of revenue. A dedicated loyalty tier — early access, free-shipping thresholds, priority support — protects the single largest revenue pool from competitive erosion.

2. Launch a targeted win-back campaign for Disengaged/Churn-Risk customers.

Time-boxed discounts or 'we miss you' offers aimed at customers past a recency threshold (300+ days) can re-engage this segment cost-effectively, since switching costs and expectations are already low.

3. Convert Occasional Shoppers into Steady Regulars with frequency incentives.

The gap versus Steady Regulars is mainly frequency, not spend-per-order — 'your 3rd purchase this quarter unlocks a reward' beats blanket discounts.

4. Instrument stronger churn signals before relying on the current label.

Since churn doesn't correlate with RFM behavior, invest in tracking real engagement signals (site visits, cart abandonment, support contacts) for genuine predictive power.

5. Personalize recommendations at the individual level, not the segment level.

Category mix barely differs by segment, so a recommendation engine keyed to each customer's own purchase history will outperform a segment-wide merchandising push.